ALGAMES

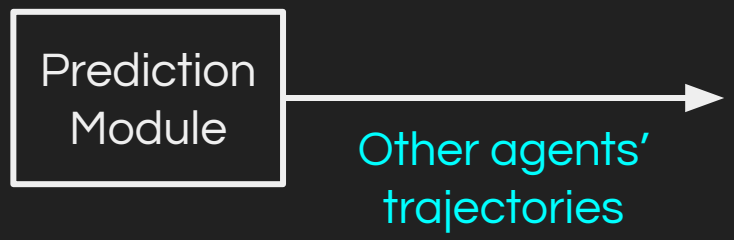
A Fast Solver for Constrained Dynamic Games

Simon Le Cleac'h , Mac Schwager and Zachary Manchester

Predict-then-plan

Prediction Module

Predict-then-plan



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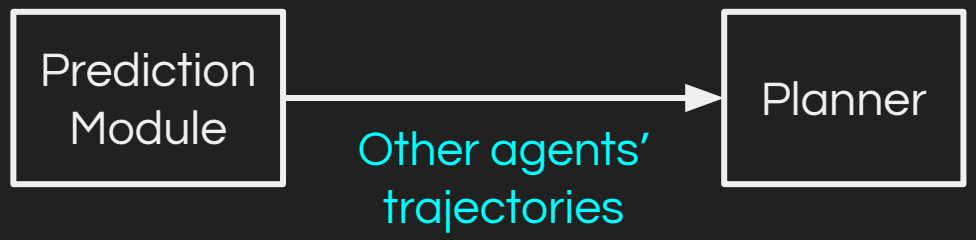

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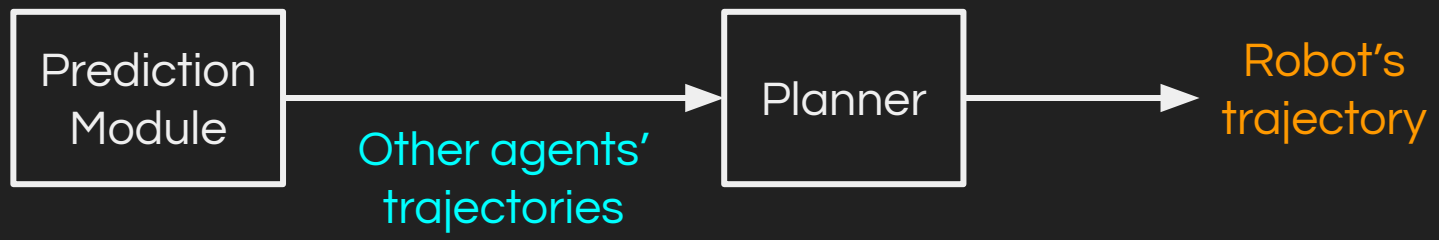
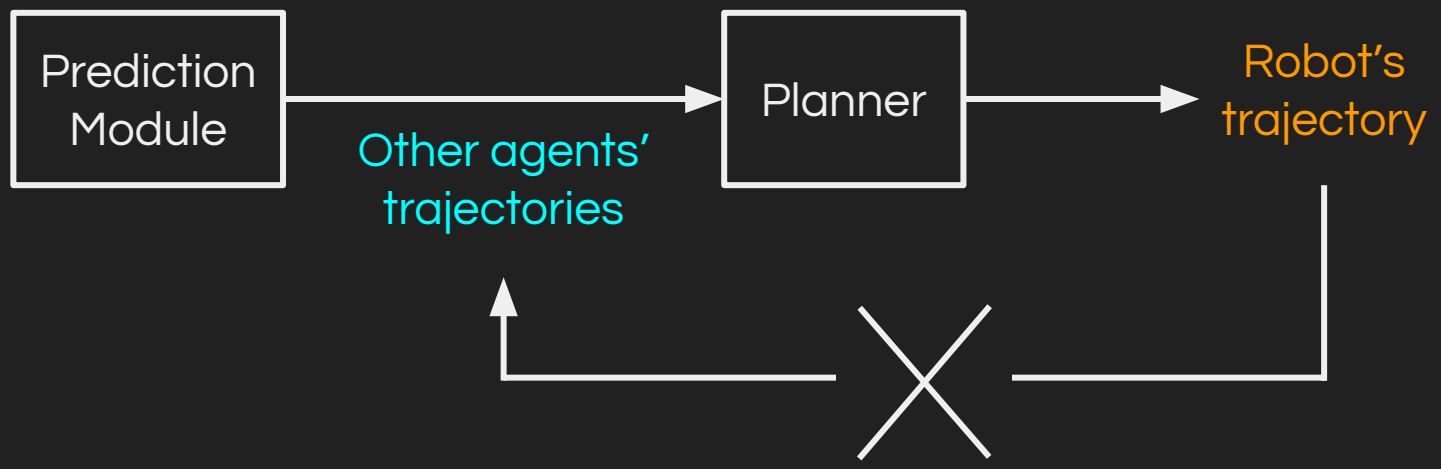


Predict-then-plan



Joint Prediction & Planning

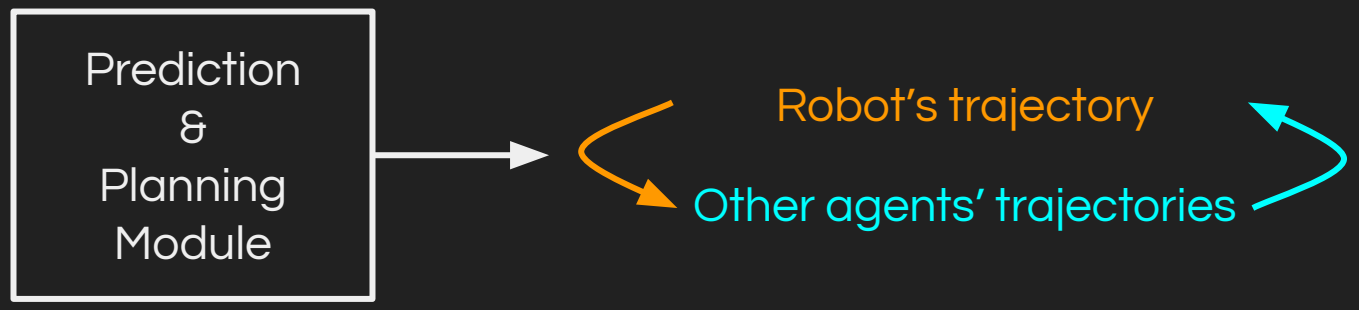
Prediction
&
Planning
Module

Joint Prediction & Planning

Robot's trajectory

Other agents' trajectories

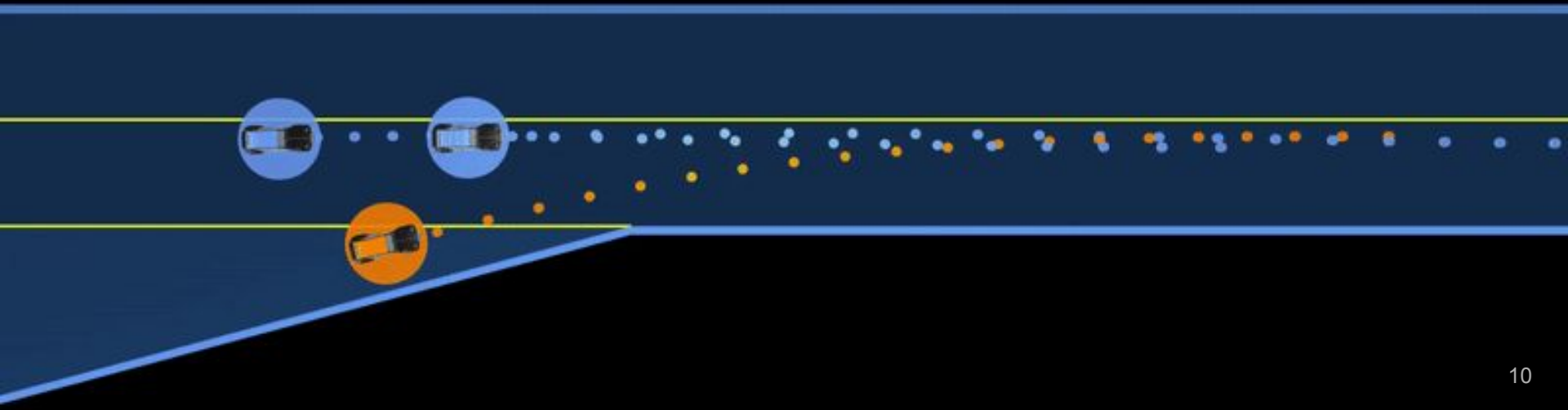
Joint Prediction & Planning



ALGAMES algorithm

Robot

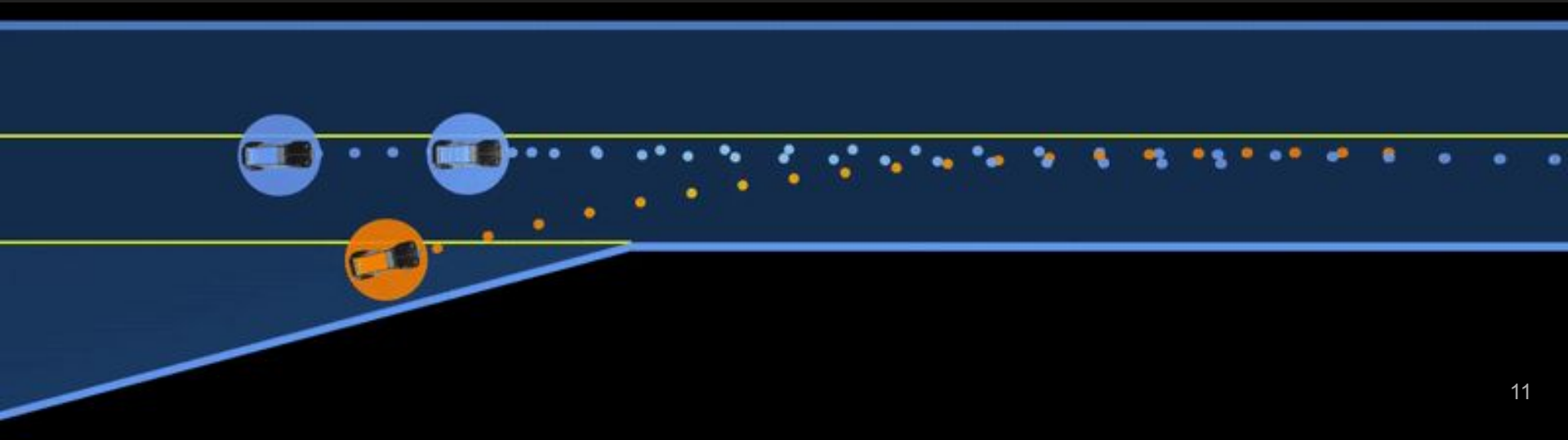
Other agents



ALGAMES algorithm

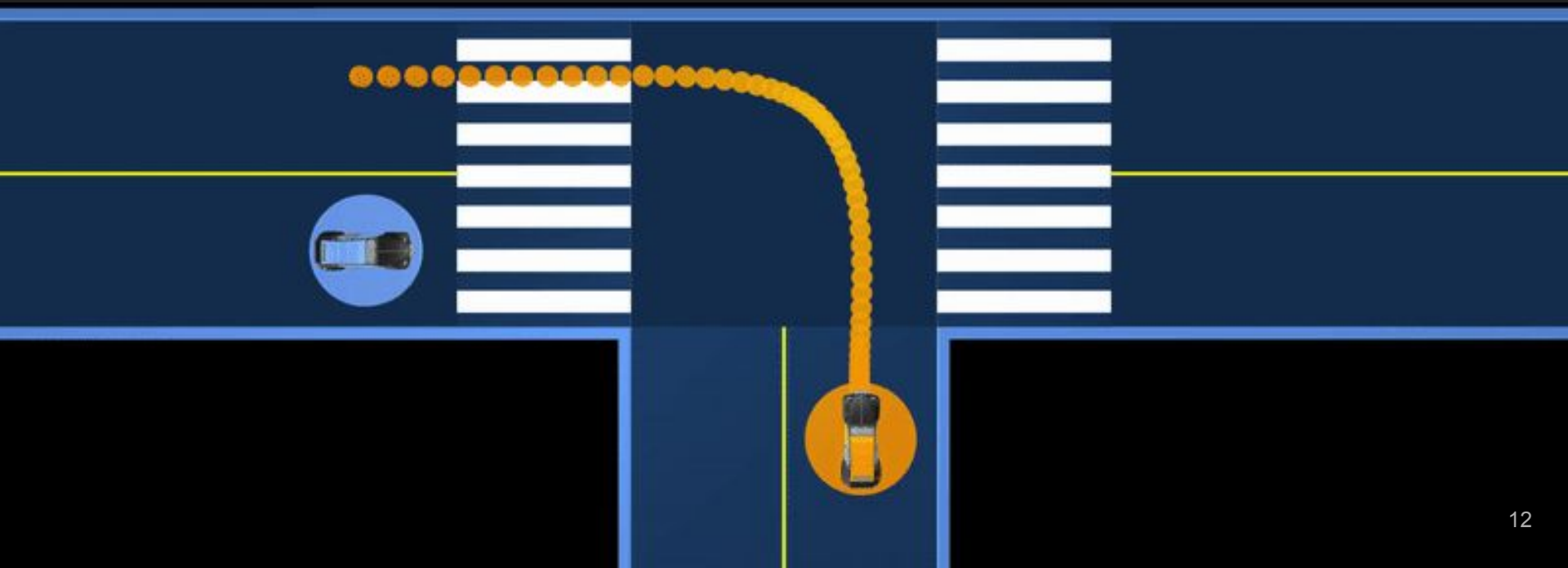
●●●● Robot's open-loop plan

●●●● Robot's trajectory predictions



Equilibrium Selection

Nash Equilibrium



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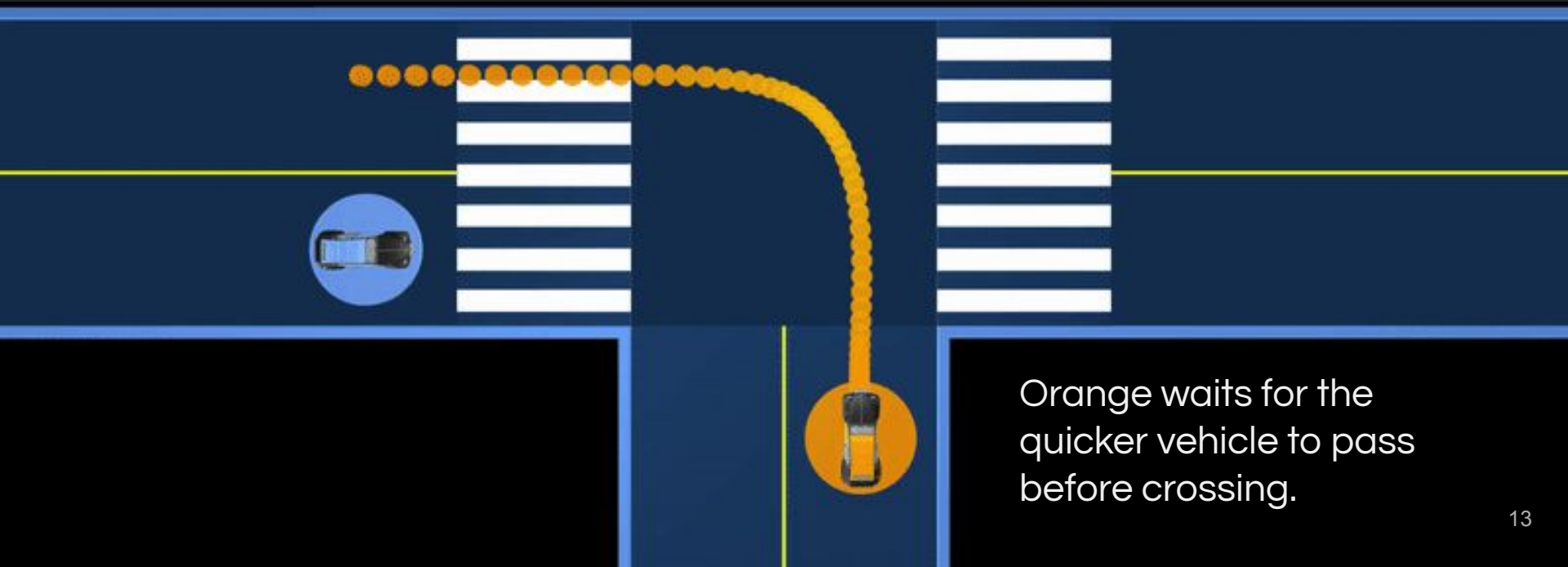
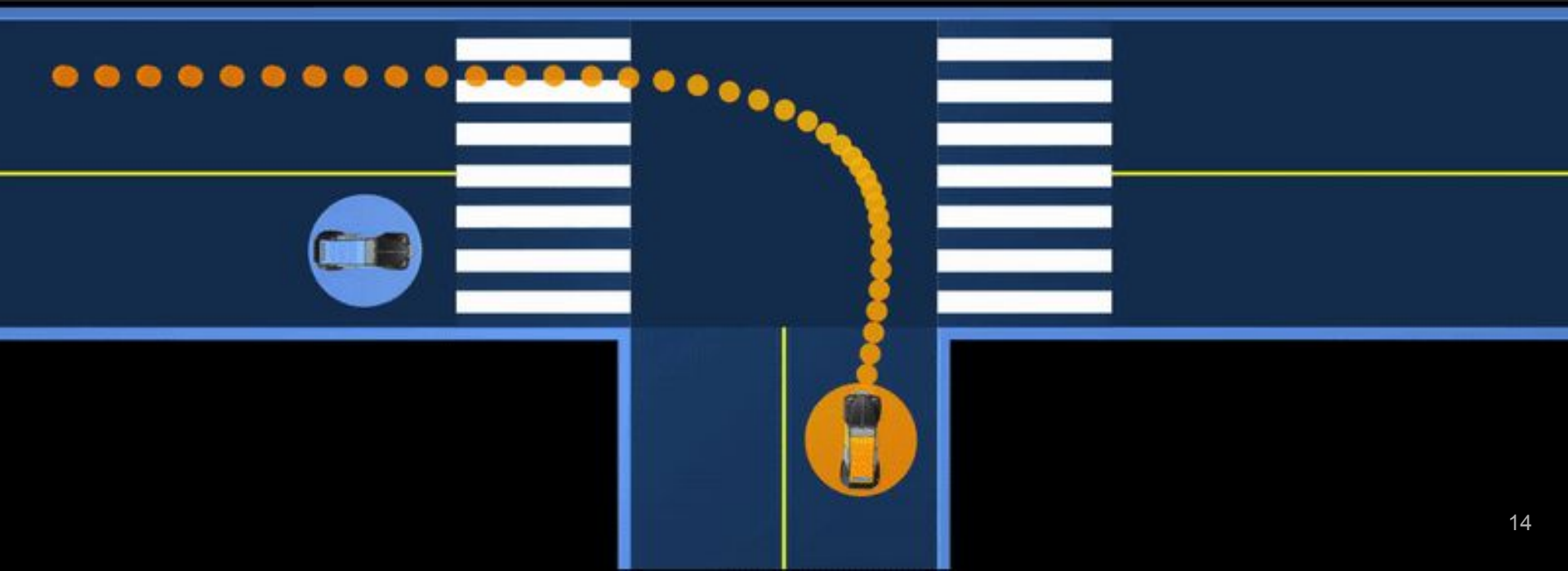


Orange waits for the quicker vehicle to pass before crossing.

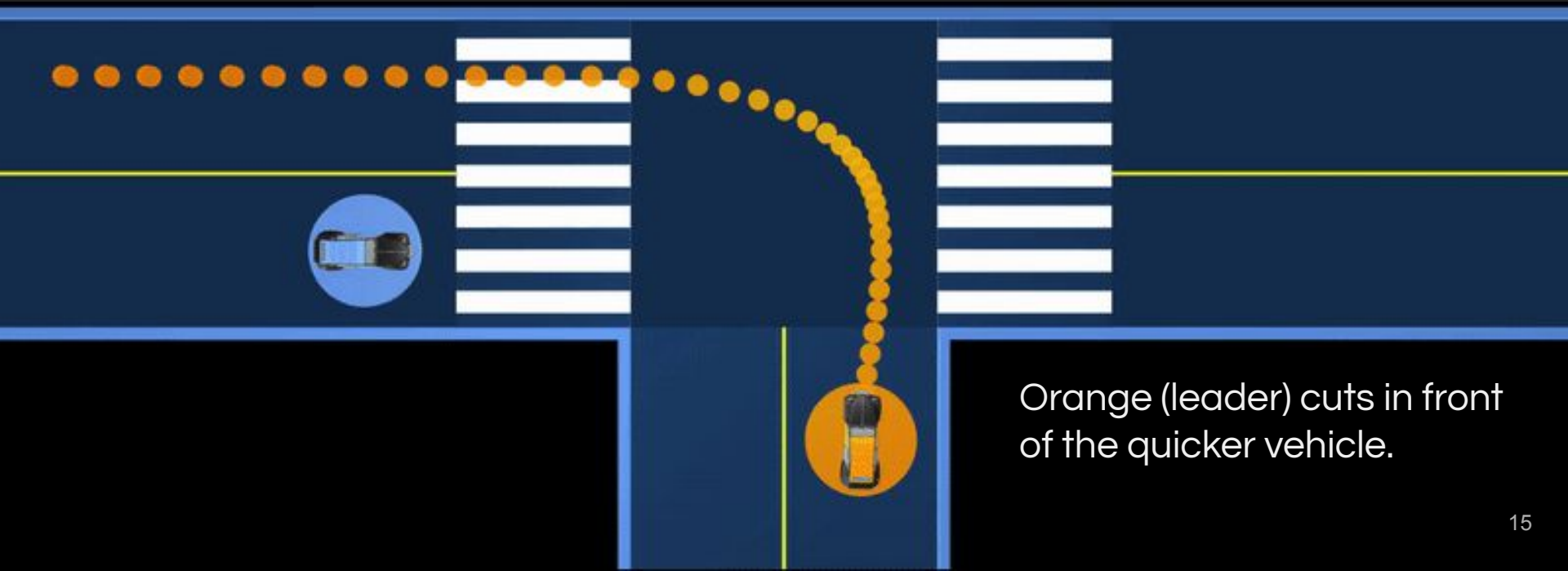
Equilibrium Selection

Stackelberg Equilibrium



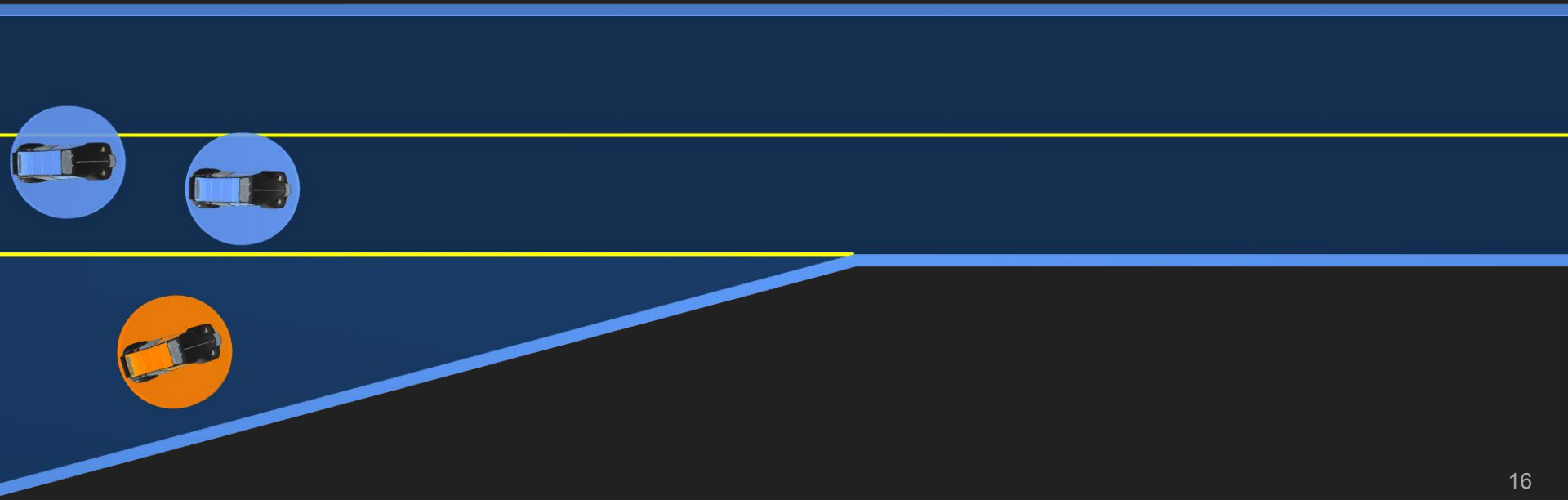
Equilibrium Selection

Stackelberg Equilibrium



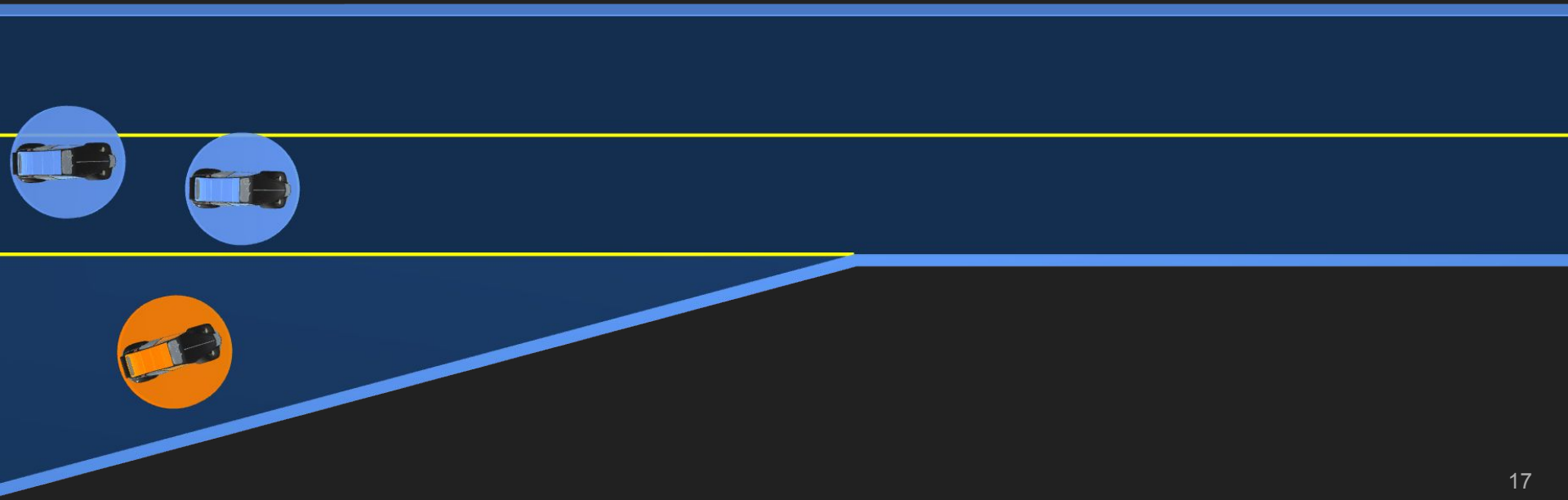
Orange (leader) cuts in front of the quicker vehicle.

ALGAMES algorithm



ALGAMES algorithm

Vehicle collision avoidance



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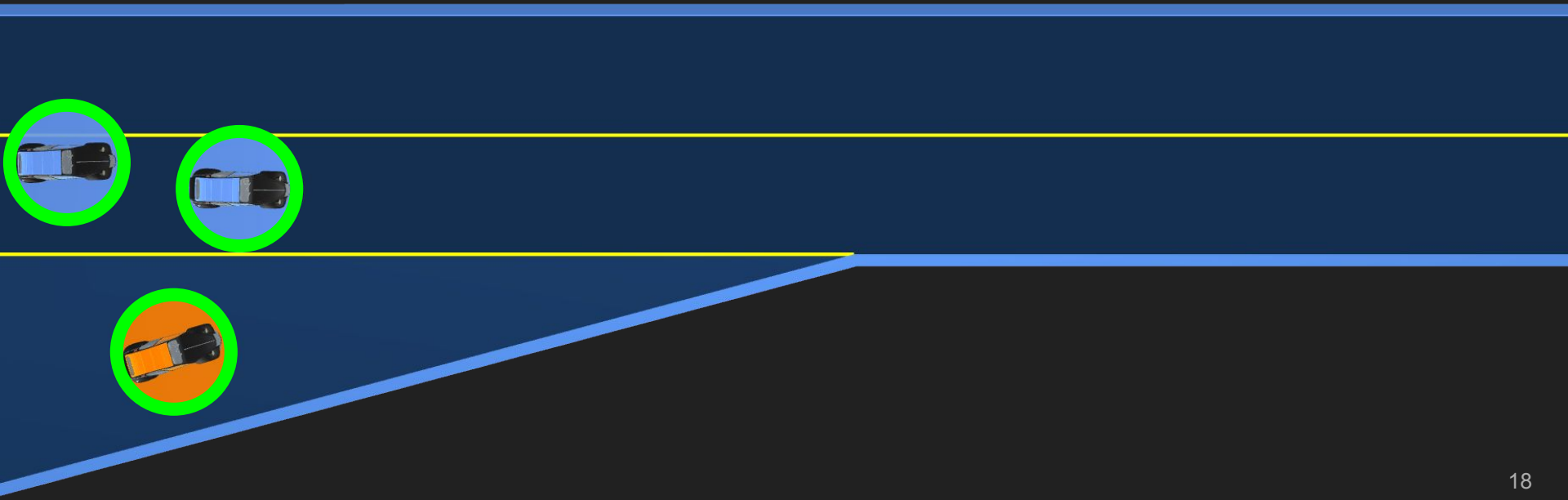
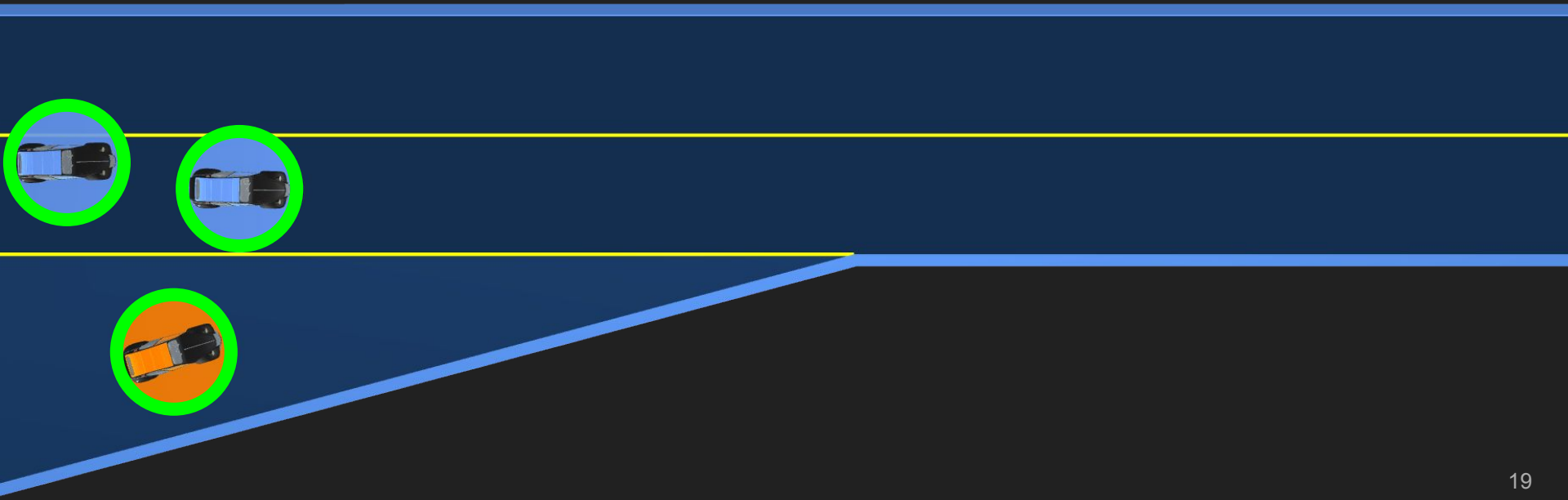


ALGAMES algorithm

Vehicle collision avoidance

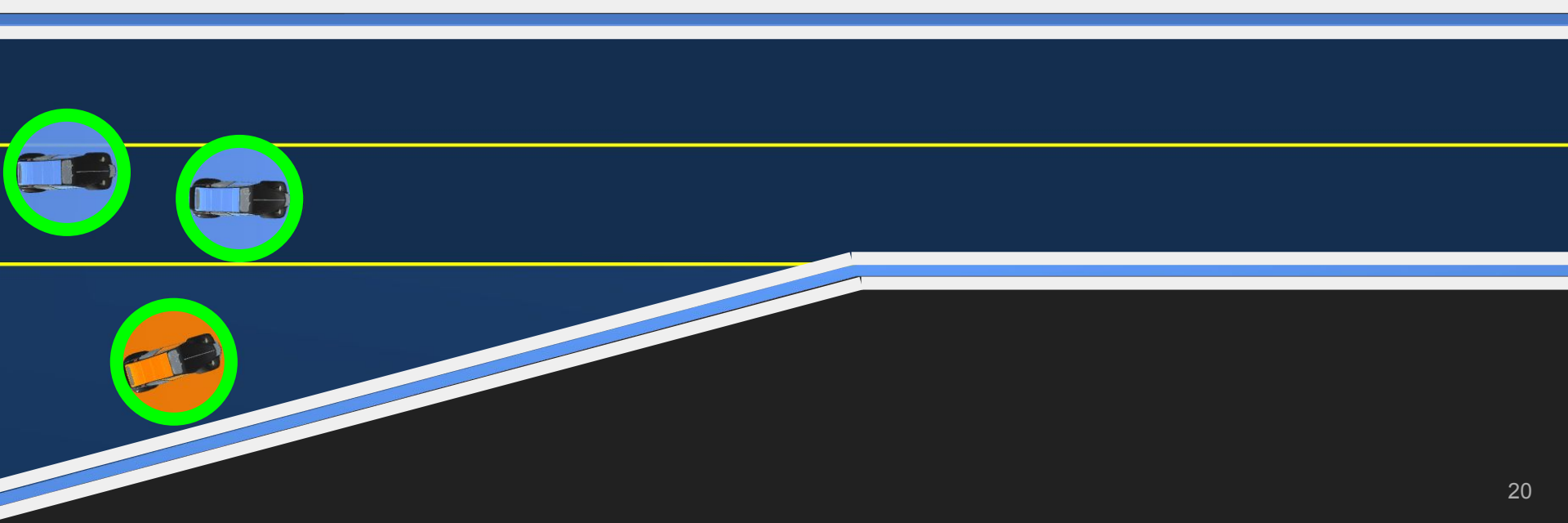
Boundary collision avoidance



ALGAMES algorithm

Vehicle collision avoidance

Boundary collision avoidance



ALGAMES algorithm

$$x_{k+1} = f(x_k, u_k^{(1)}, u_k^{(2)})$$

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ALGAMES algorithm

$$\min_{X, U^{(1)}} \quad J^1(X, U^{(1)}),$$

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$$\min_{X, U^{(1)}} \quad \overbrace{J^1}^{\text{Cost}}(X, U^{(1)}),$$

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$$\min_{X, U^{(1)}} \quad \overbrace{J^1}^{\text{Cost}}(\overbrace{X, U^{(1)}}^{\text{State \& Control Sequences}}),$$

$$\text{s.t.} \quad C(X, U) \leq 0.$$

ALGAMES algorithm

$$\min_{X, U^{(1)}} \quad \overbrace{J^1}^{\text{Cost}}(\overbrace{X, U^{(1)}}^{\text{State \& Control Sequences}}),$$

$$\text{s.t.} \quad \underbrace{C(X, U)}_{\text{Constraints: dynamics, collisions etc.}} \leq 0.$$

ALGAMES algorithm

Agent 1

$$\left\{ \begin{array}{ll} \min\limits_{X, U^{(1)}} & J^1(X, U^{(1)}), \\ \text{s.t.} & C(X, U) \leq 0. \end{array} \right\}$$

ALGAMES algorithm

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$$\left\{\begin{array}{ll} \min\limits_{X, U^{(1)}} & J^1(X, U^{(1)}), \\ \text{s.t.} & C(X, U) \leq 0. \end{array}\right\}$$

Agent 2

$$\left\{\begin{array}{ll} \min\limits_{X, U^{(2)}} & J^2(X, U^{(2)}), \\ \text{s.t.} & C(X, U) \leq 0. \end{array}\right\}$$

ALGAMES algorithm

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Augmented Lagrangian

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Augmented Lagrangian

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Augmented Lagrangian

Optimality

Agent 2

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ALGAMES algorithm

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Augmented Lagrangian

Optimality

Gauss Newton

Agent 2

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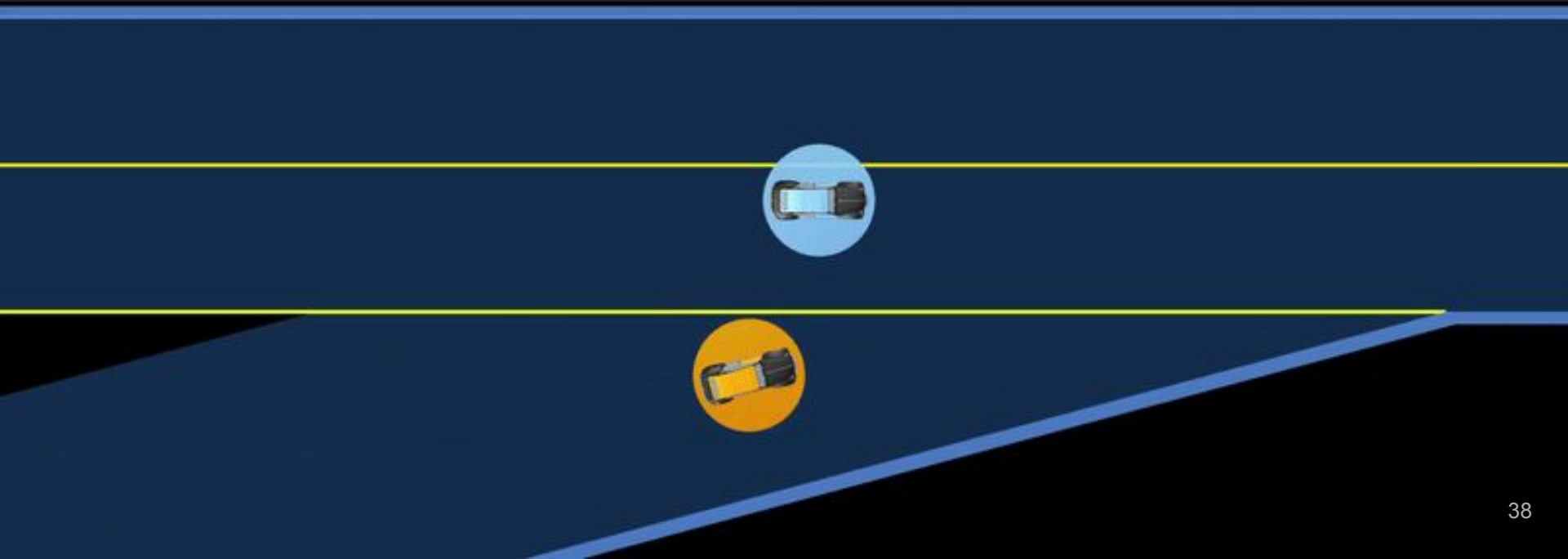
Applications

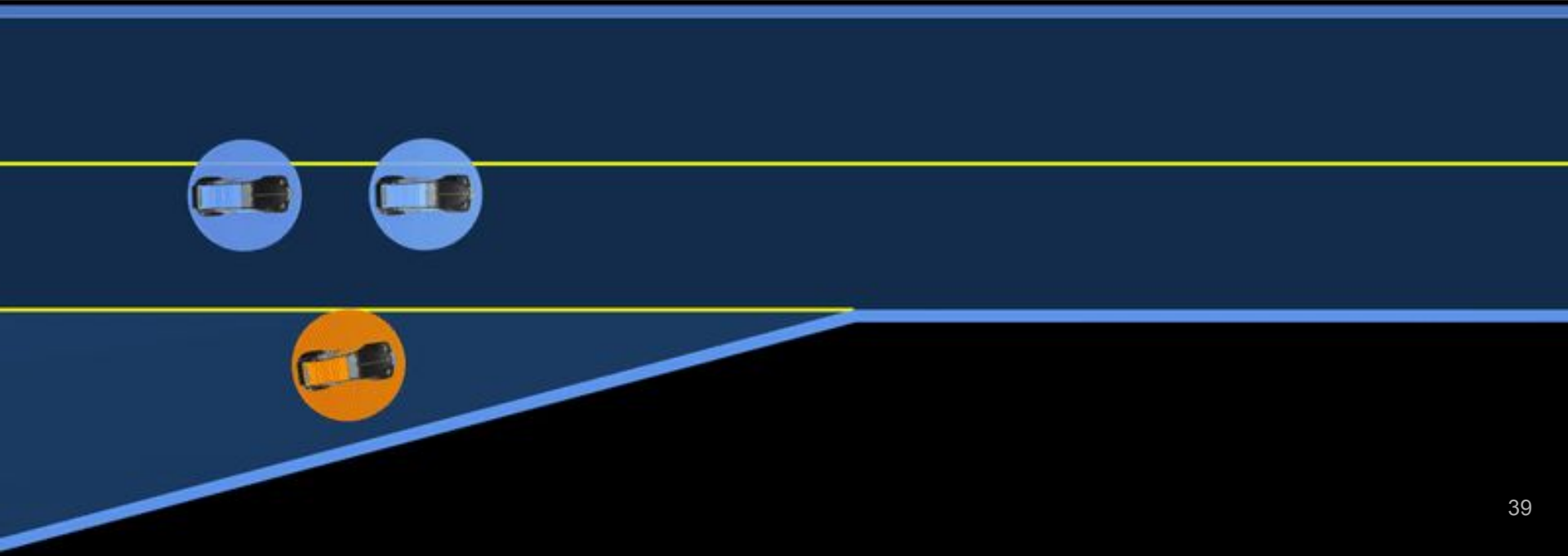
Applications

Lane changing



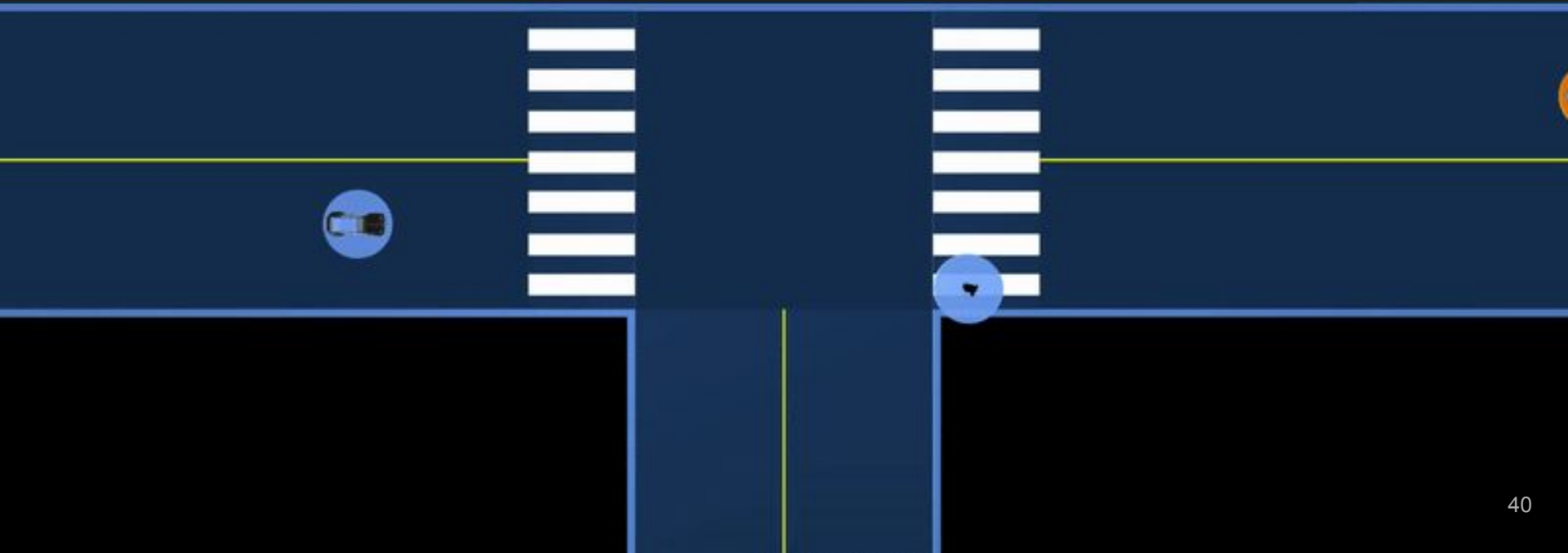
Applications

Ramp merging

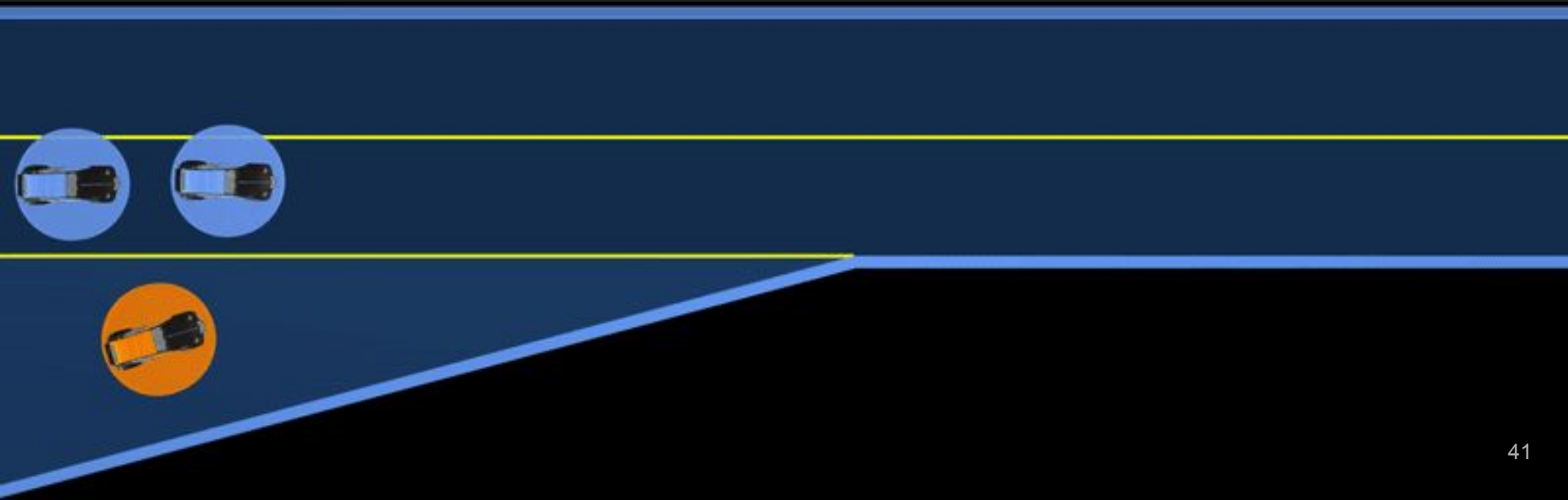


Applications

Intersection crossing

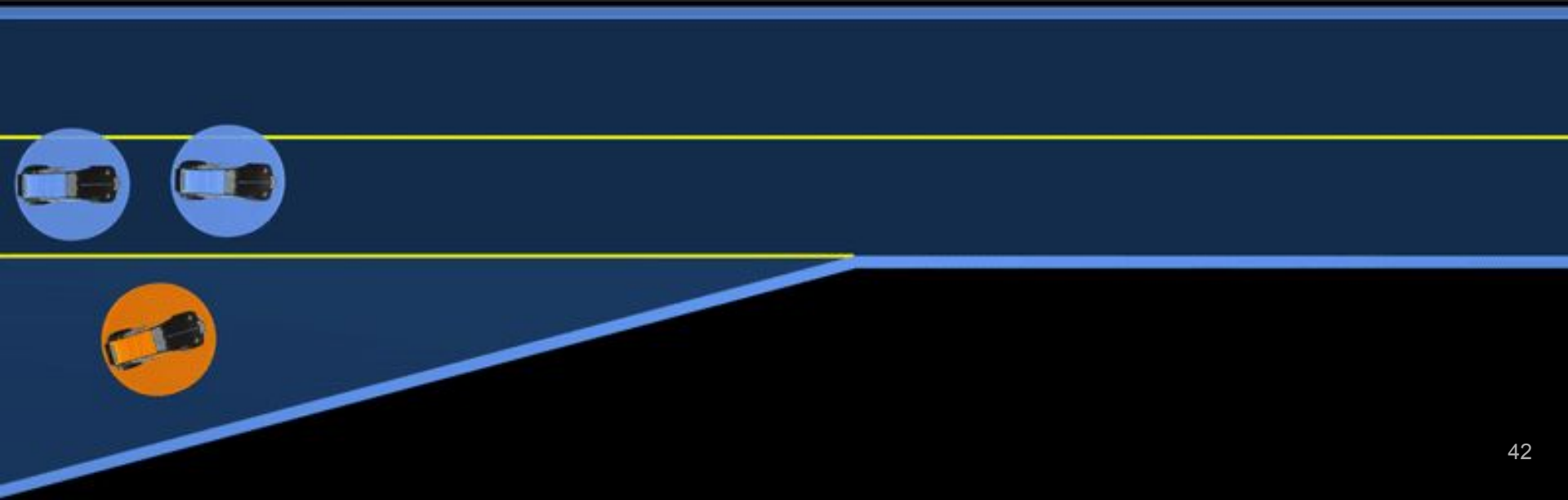


Speed and Robustness



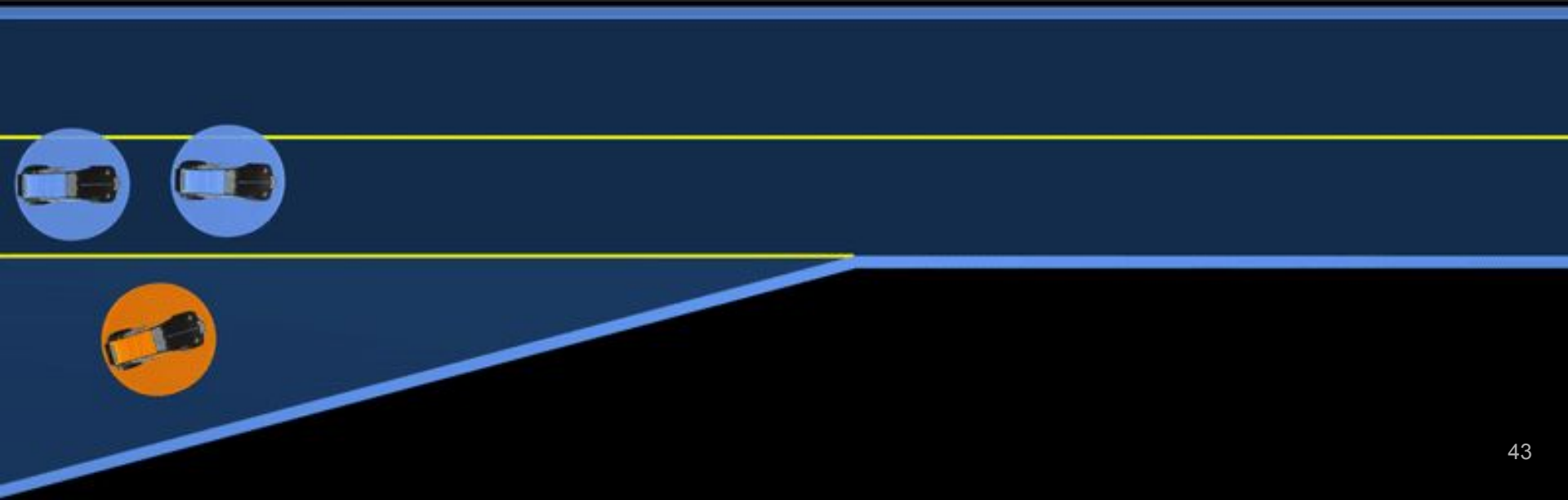
Speed and Robustness

- ALGAMES: 3 times faster than iLQGames.

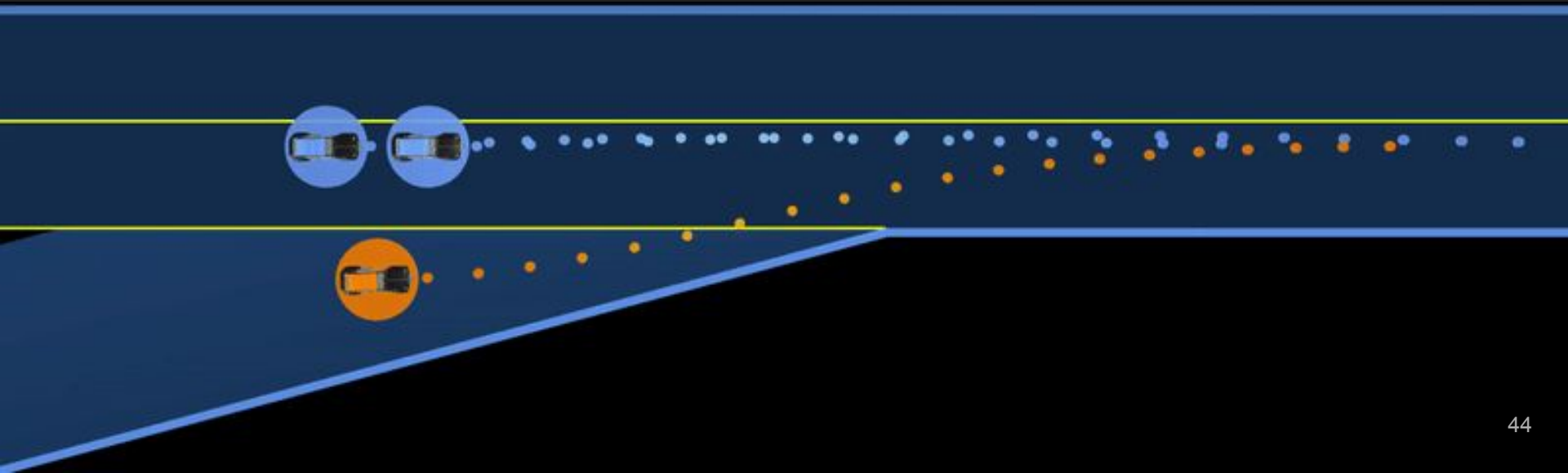


Speed and Robustness

- ALGAMES: 3 times faster than iLQGames.
- 99.5% successful solves.

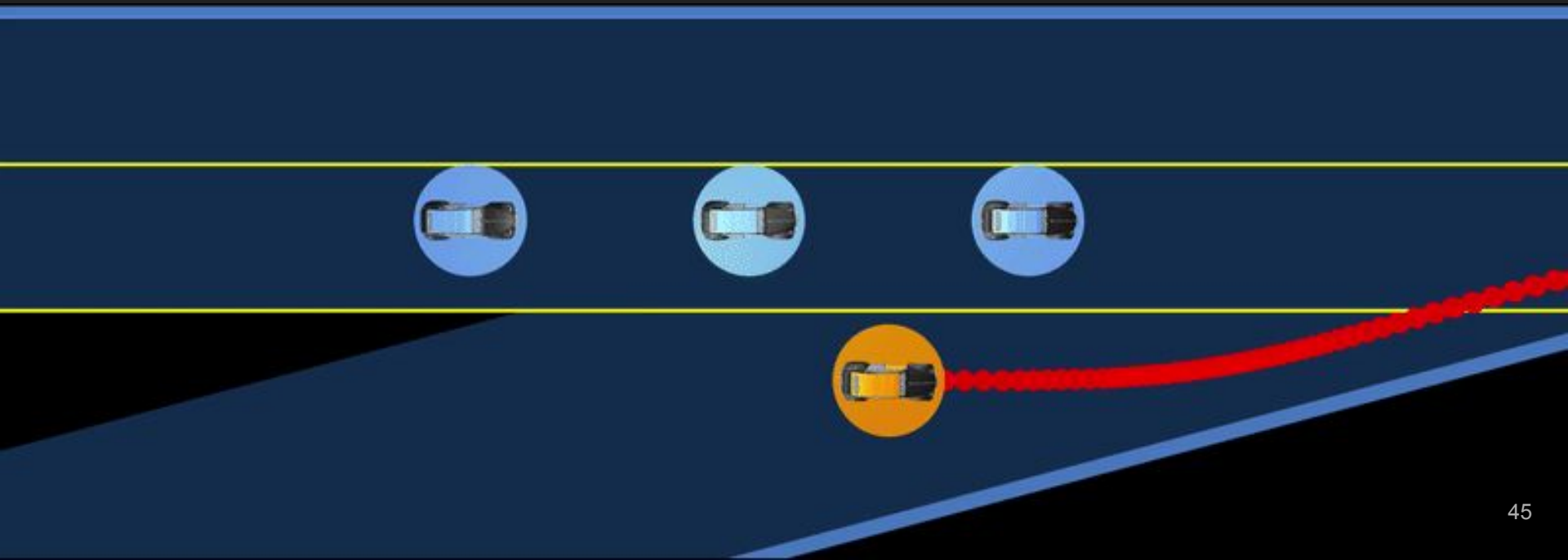


Model Predictive Control



Model Predictive Control

Non-game theoretic MPC



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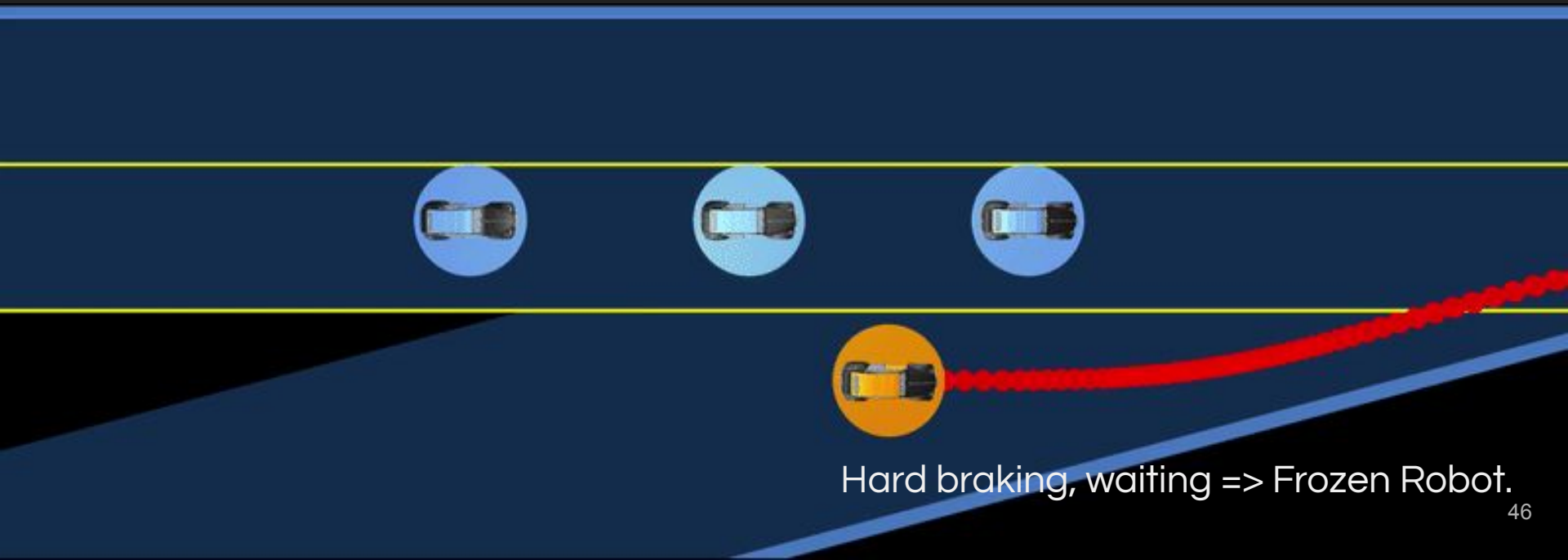
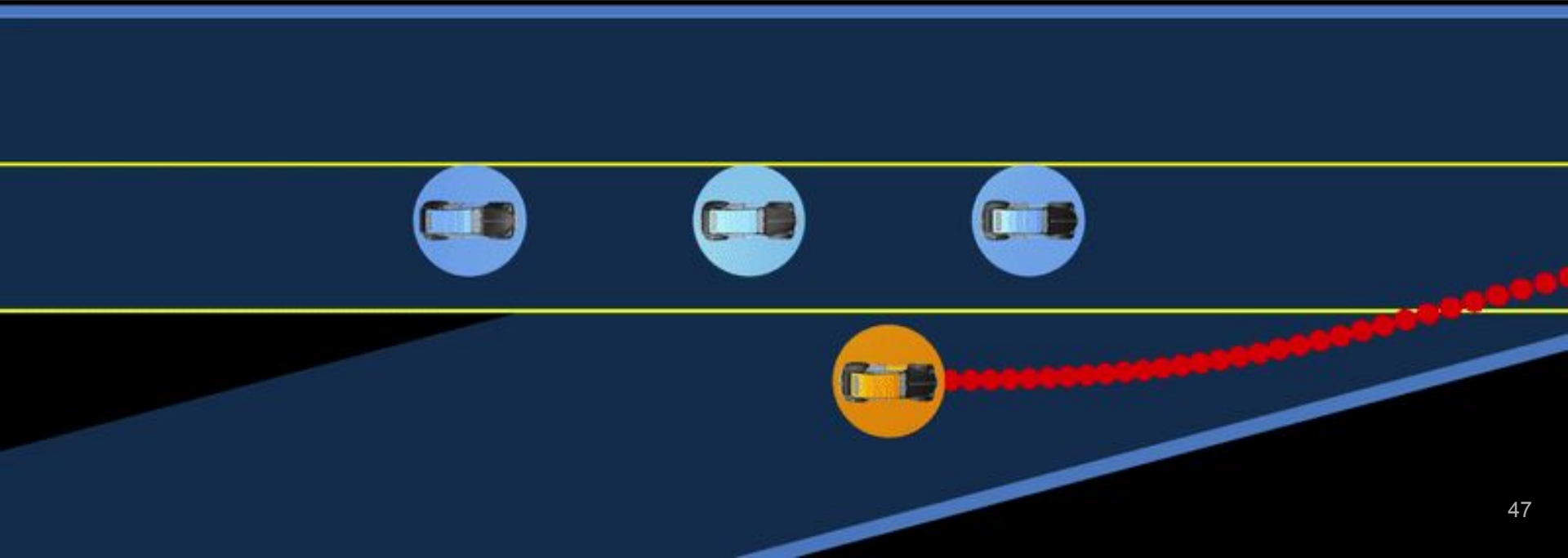


Hard braking, waiting => Frozen Robot.

Model Predictive Control

ALGAMES MPC



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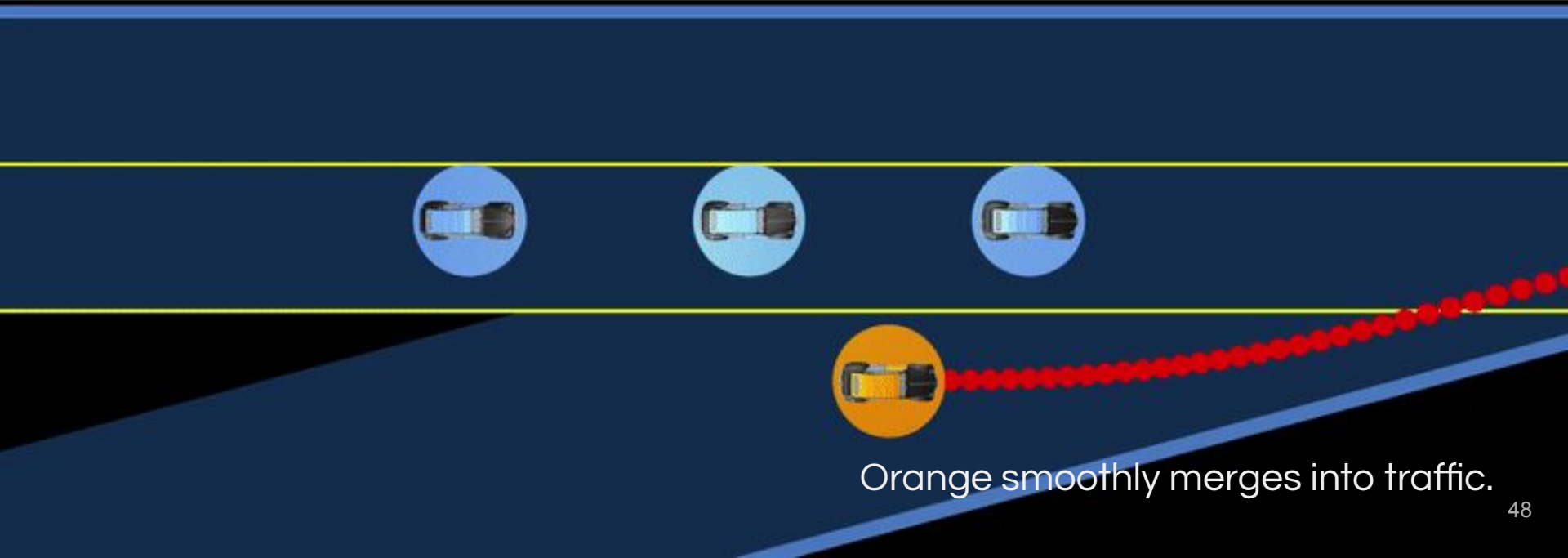

Orange smoothly merges into traffic.

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https://github.com/RoboticExplorationLab/ALGAMES.jl.git

Questions?